# CSE 150A-250A AI: Probabilistic Models

## Lecture 9

Fall 2026

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Slides adapted from previous versions of the course (Prof. Lawrence, Prof. Alvarado, Prof Berg-Kirkpatrick)

# Agenda

Review

Incomplete Data

Expectation-Maximization Algorithm

# Review

# Learning in BNs with discrete nodes

- ML estimation for complete data:

$$P_{\mathrm{ML}}(X_i = x | \mathrm{pa}_i = \pi) = \frac{\mathrm{count}(X_i = x, \mathrm{pa}_i = \pi)}{\sum_{x'} \mathrm{count}(X_i = x', \mathrm{pa}_i = \pi)}$$

CPT

- For nodes with parents:

$$P_{\mathrm{ML}}(X_i = x | \mathrm{pa}_i = \pi) = \frac{\mathrm{count}(X_i = x, \mathrm{pa}_i = \pi)}{\mathrm{count}(\mathrm{pa}_i = \pi)}$$

- For root nodes:

$$P_{\mathrm{ML}}(X_i = x) = \frac{\mathrm{count}(X_i = x)}{T}$$

# Markov models for statistical language processing

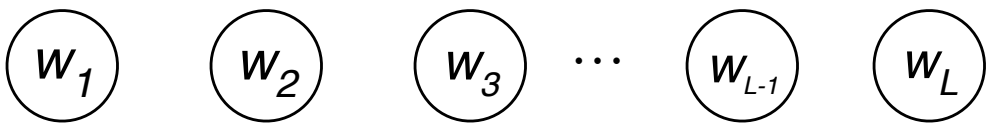
- $n$-gram models of word sequences:

$$P(w_1, w_2, \ldots, w_L) = \prod_{\ell} P(w_\ell | \underbrace{w_{\ell-(n-1}, \ldots, w_{\ell-1})}_{\text{previous words}})$$

- As belief networks:

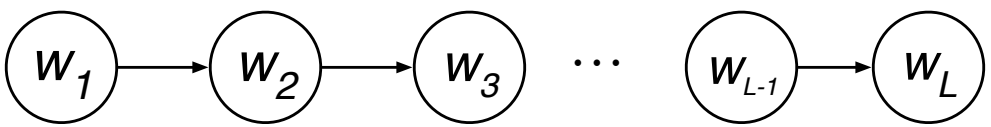
$n = 1$

unigram



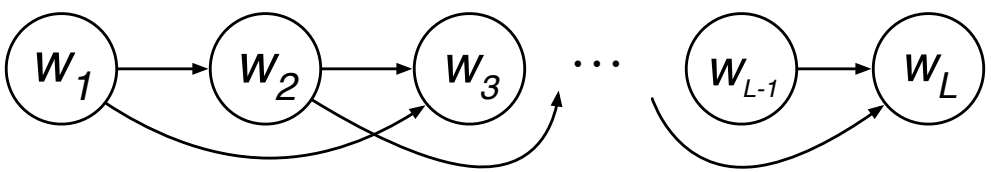

$n = 2$

bigram



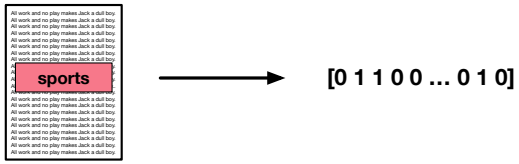

$n = 3$

trigram



# Naive Bayes model for document classification

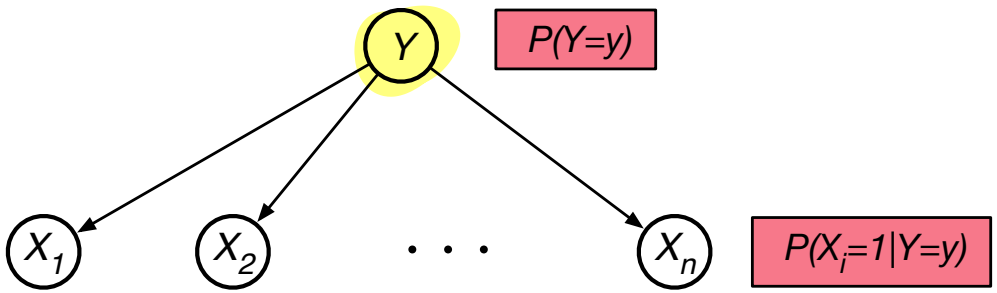
- Random variables




$Y \in \{1, 2, \ldots, m\}$ topic of document

$X_i \in \{0, 1\}$ $i^{\text{th}}$ word appears?

- Belief network




New doc
P(Y|w...)
=

- Naive Bayes assumption

$$P(X_1, \ldots, X_n | Y) = \prod_{i=1}^{n} P(X_i | Y)$$

# Incomplete Data

# Learning from incomplete data with tabular CPTs

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ASSUMPTIONS

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1. The DAG is fixed (and known) over a finite set of discrete random variables $\{X_1, X_2, \ldots, X_n\}$.

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2. CPTs enumerate $P(X_i = x | \text{pa}(X_i) = \pi)$ as lookup tables; each must be estimated for all values of $x$ and $\pi$.

# Learning from incomplete data with tabular CPTs

ASSUMPTIONS

1. The DAG is fixed (and known) over a finite set of discrete random variables $\{X_1, X_2, \ldots, X_n\}$.

2. CPTs enumerate $P(X_i = x | \text{pa}(X_i) = \pi)$ as lookup tables; each must be estimated for all values of $x$ and $\pi$.

3. The data is IID, but only consists of $T$ partially complete instantiations of the nodes in the BN.

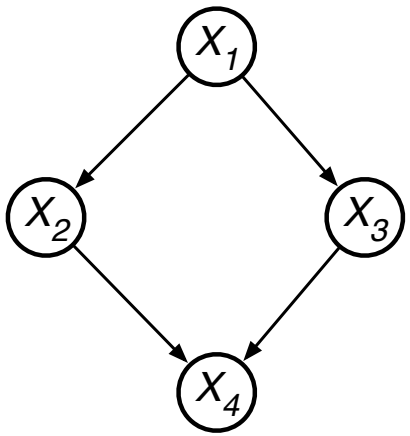
# Toy example

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- Fixed DAG over binary random variables

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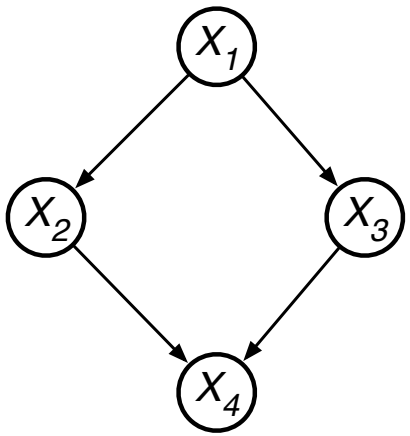
- Fixed DAG over binary random variables




$$
\begin{aligned}
X_1 &\in \{0,1\} \\
X_2 &\in \{0,1\} \\
X_3 &\in \{0,1\} \\
X_4 &\in \{0,1\}
\end{aligned}
$$

# Toy example

- Fixed DAG over binary random variables

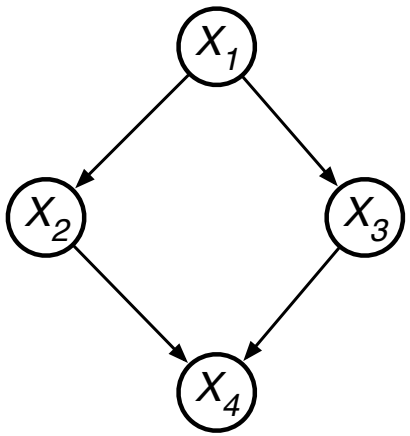



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$$

- Incomplete data set

# Toy example

- Fixed DAG over binary random variables



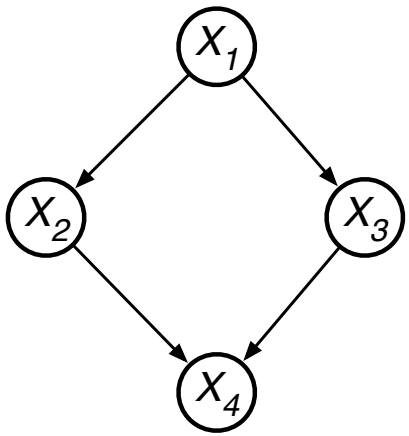

$$X_1 \in \{0,1\}$$
$$X_2 \in \{0,1\}$$
$$X_3 \in \{0,1\}$$
$$X_4 \in \{0,1\}$$

- Incomplete data set

| example | $X_1$ | $X_2$ | $X_3$ | $X_4$ |
|---|---|---|---|---|
| 1 | 1 | ? | 0 | 1 |
| 2 | 0 | 1 | ? | 0 |
| 3 | ? | ? | ? | 1 |
| : | : | : | : | : |
| T | ? | 1 | 1 | 0 |

# Toy example

- Fixed DAG over binary random variables




$$X_1 \in \{0,1\}$$
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| : | : | : | : | : |
| T | ? | 1 | 1 | 0 |

How to choose the CPTs so that the BN maximizes the probability of this data set?

# A more interesting example ...

# A more interesting example ...





















# A more interesting example ...
















How to build a movie recommendation system?

# A more interesting example ...
















How to build a movie recommendation system?

- Collect a data set of movie ratings:

# A more interesting example ...















**How to build a movie recommendation system?**

- Collect a data set of movie ratings:

$$\begin{bmatrix} + & - & + & - & ? & ? & + \\ - & ? & ? & + & + & ? & ? \\ + & + & + & + & + & + & + \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ - & - & - & - & - & ? & - \\ ? & ? & + & ? & ? & ? & - \end{bmatrix}$$

| | |
|---|---|
| $+$ | liked |
| $-$ | disliked |
| ? | not seen |

(user-item matrix)

# A more interesting example ...
















**How to build a movie recommendation system?**

- Collect a data set of movie ratings:

$$\begin{bmatrix} + & - & + & - & ? & ? & + \\ - & ? & ? & + & + & ? & ? \\ + & + & + & + & + & + & + \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ - & - & - & - & - & ? & - \\ ? & ? & + & ? & ? & ? & - \end{bmatrix}$$

| | |
|---|---|
| $+$ | liked |
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**(user-item matrix)**

- Build a model of user profiles and fill in the missing ratings.
  But what model to build?

# Naive Bayes model with incomplete data

# Naive Bayes model with incomplete data





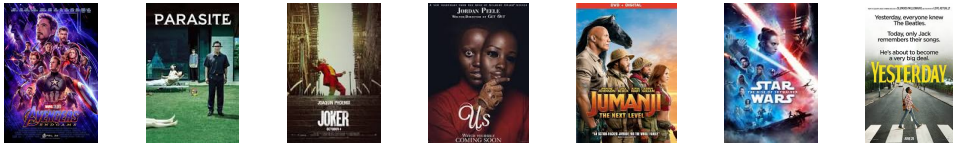


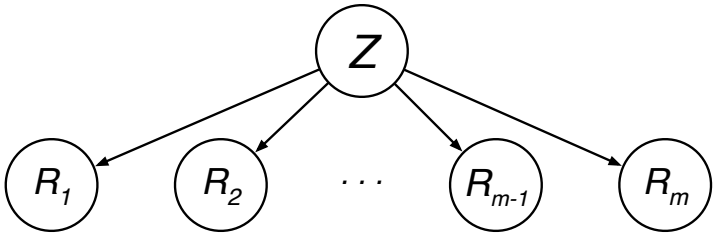




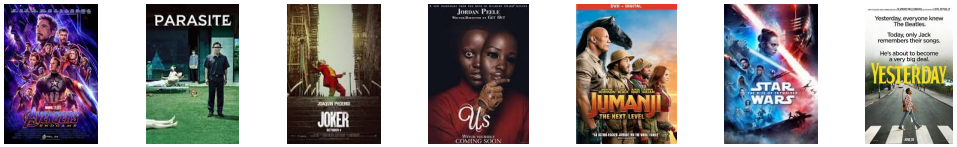


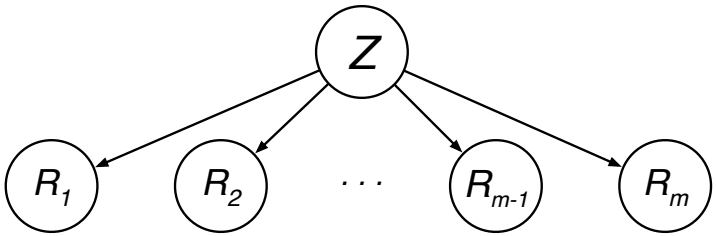

# Naive Bayes model with incomplete data



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- Movie recommender system

# Naive Bayes model with incomplete data





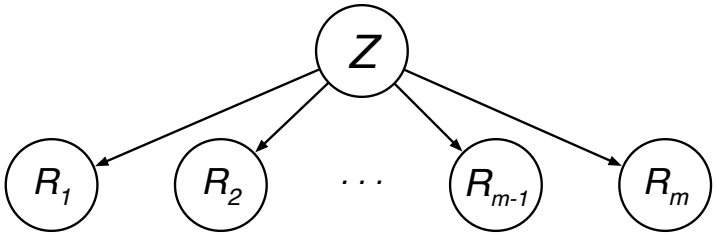








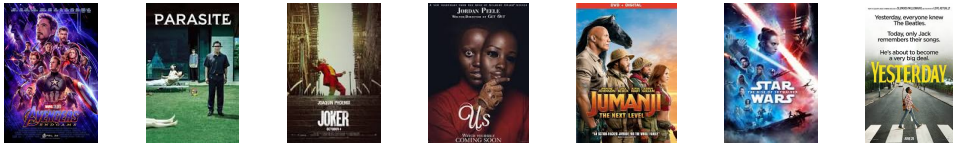


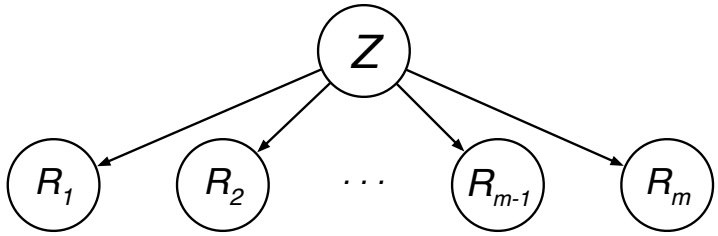



- Movie recommender system

$$Z \in \{1, 2, \ldots, k\} \quad \text{type of movie-goer}$$

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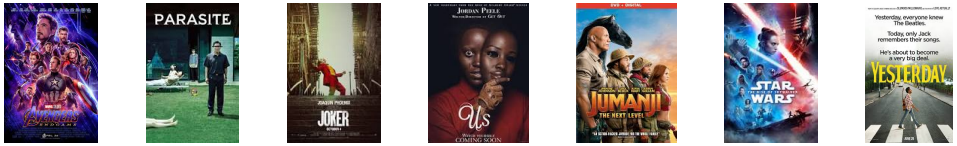


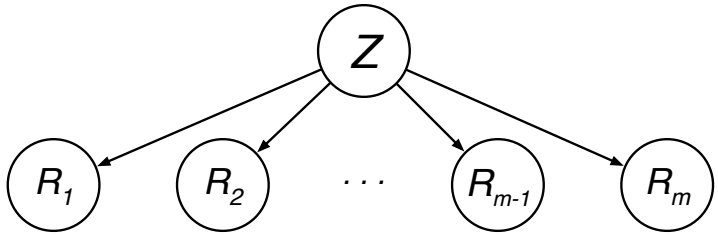




- Movie recommender system

$$
\begin{aligned}
Z &\in \{1, 2, \ldots, k\} && \text{type of movie-goer} \\
R_i &\in \{0, 1\} && \text{rating for } i^{\text{th}} \text{ movie}
\end{aligned}
$$

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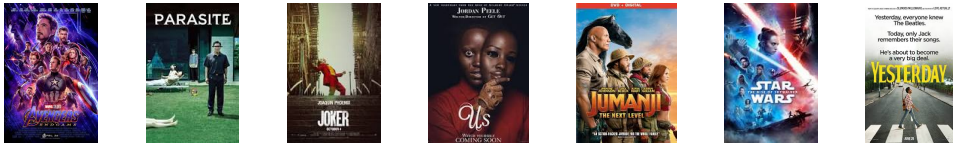



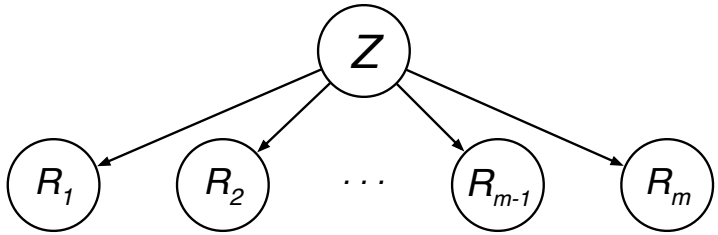

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# Naive Bayes model with incomplete data



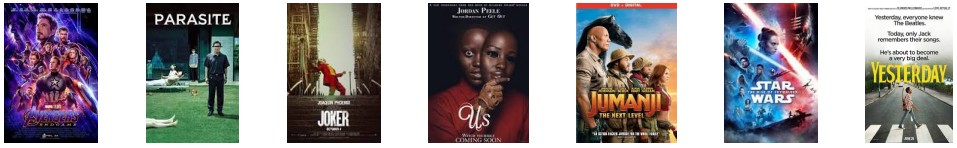

- Movie recommender system

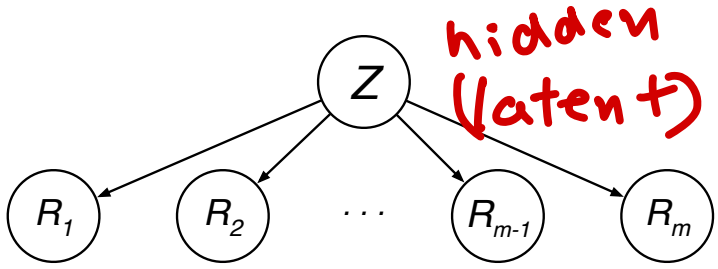
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Z &\in \{1, 2, \ldots, k\} && \text{type of movie-goer} \\
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$$

- Incomplete data set

| student | $Z$ | $R_1$ | $R_2$ | $R_3$ | $R_4$ | $\cdots$ |
|---|---|---|---|---|---|---|
| 1 | ? | 0 | 1 | 1 | ? | $\cdots$ |
| 2 | ? | 1 | ? | 0 | 1 | $\cdots$ |
| 3 | ? | 0 | 0 | ? | 1 | $\cdots$ |
| : | : | : | : | : | : | : |
| T | ? | ? | 1 | 0 | ? | $\cdots$ |

# Naive Bayes model with incomplete data




- Movie recommender system

$$Z \in \{1, 2, \ldots, k\} \quad \text{type of movie-goer}$$
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- Incomplete data set

| student | $Z$ | $R_1$ | $R_2$ | $R_3$ | $R_4$ | ... |
|---|---|---|---|---|---|---|
| 1 | ? | 0 | 1 | 1 | ? | ... |
| 2 | ? | 1 | ? | 0 | 1 | ... |
| 3 | ? | 0 | 0 | ? | 1 | ... |
| : | : | : | : | : | : | : |
| T | ? | ? | 1 | 0 | ? | ... |

Note that the variable $Z$ is never observed.

# Learning from incomplete data

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$H_t$ = set of hidden (latent) variables for $t^{\text{th}}$ example

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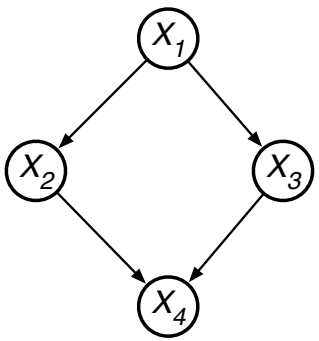
- Illustration

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- Illustration

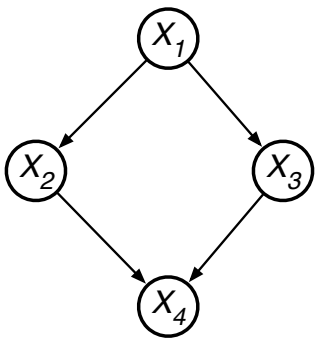


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- Illustration




| example | $X_1$ | $X_2$ | $X_3$ | $X_4$ |
|---|---|---|---|---|
| 1 | 1 | ? | 0 | 1 |
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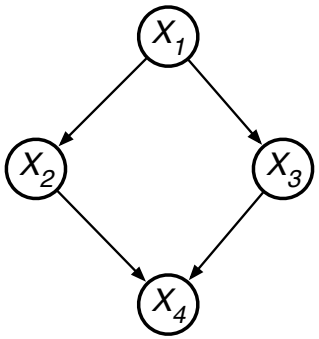
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| example | $X_1$ | $X_2$ | $X_3$ | $X_4$ |
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| : | : | : | : | : |

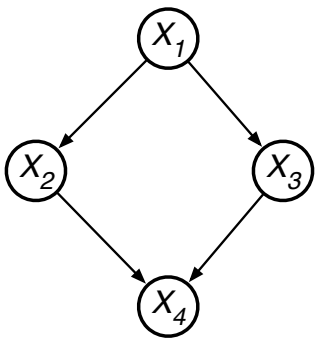
$H_1 = \{X_2\}$ $\quad$ $V_1 = \{X_1, X_3, X_4\}$

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$$H_t = \text{set of hidden (latent) variables for } t^{\text{th}} \text{ example}$$
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- Illustration




| example | $X_1$ | $X_2$ | $X_3$ | $X_4$ |
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| : | : | : | : | : |

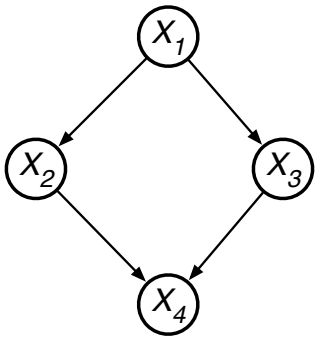
$$H_1 = \{X_2\} \qquad V_1 = \{X_1, X_3, X_4\}$$
$$H_2 = \{X_3\} \qquad V_2 = \{X_1, X_2, X_4\}$$

# Learning from incomplete data

- Notation

$$H_t = \text{set of hidden (latent) variables for } t^{\text{th}} \text{ example}$$
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- Illustration




| example | $X_1$ | $X_2$ | $X_3$ | $X_4$ |
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| : | : | : | : | : |

$$H_1 = \{X_2\} \qquad V_1 = \{X_1, X_3, X_4\}$$
$$H_2 = \{X_3\} \qquad V_2 = \{X_1, X_2, X_4\}$$
$$H_3 = \{X_1, X_2, X_3\} \qquad V_3 = \{X_4\}$$

# Computing the log-likelihood with incomplete data

# Computing the log-likelihood with incomplete data

$$\mathcal{L} = \log P(\text{data})$$

# Computing the log-likelihood with incomplete data

$$\begin{aligned} \mathcal{L} &= \log P(\text{data}) \\ &= \log \prod_{t=1}^{T} P\left(V_t = v_t\right) \end{aligned}$$

data is IID

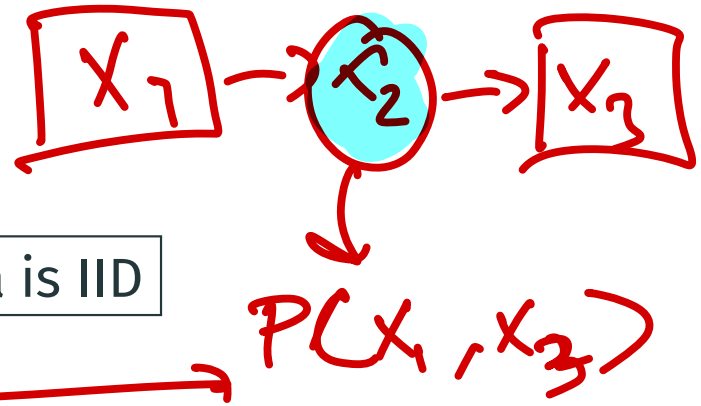
# Computing the log-likelihood with incomplete data

$$
\begin{aligned}
\mathcal{L} &= \log P(\text{data}) \\
&= \log \prod_{t=1}^{T} P\left(V_t = v_t\right) && \boxed{\text{data is IID}} \\
&= \sum_{t=1}^{T} \log P\left(V_t = v_t\right) && \boxed{\log ab = \log a + \log b}
\end{aligned}
$$

→ this is not the joint of the BN

# Computing the log-likelihood with incomplete data

$$\begin{aligned}
\mathcal{L} &= \log P(\text{data}) \\
&= \log \prod_{t=1}^{T} P(V_t = v_t) \qquad \boxed{\text{data is IID}} \\
&= \sum_{t=1}^{T} \log P(V_t = v_t) \qquad \boxed{\log ab = \log a + \log b}
\end{aligned}$$




Q. What should we do next? (to express the full joint)

A. Use product rule

B. Express $P(V_t = v_t)$ using conditional independence

C. Use marginalization

D. Use Bayes Rule

E. None of them

# Computing the log-likelihood with incomplete data

$$
\begin{aligned}
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&= \log \prod_{t=1}^{T} P\left(V_t = v_t\right) && \boxed{\text{data is IID}} \\
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The CPTs are potentially all coupled.

How to proceed?

# Expectation-Maximization Algorithm

# EM algorithm in a nutshell

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- If only the data weren't incomplete ...

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| student | $Z$ | $R_1$ | $R_2$ | $\cdots$ |
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If the data were complete, we could easily estimate the CPTs. What can we do instead?

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**Amazingly, this is how EM works (more or less) ...**

# EM algorithm — overview

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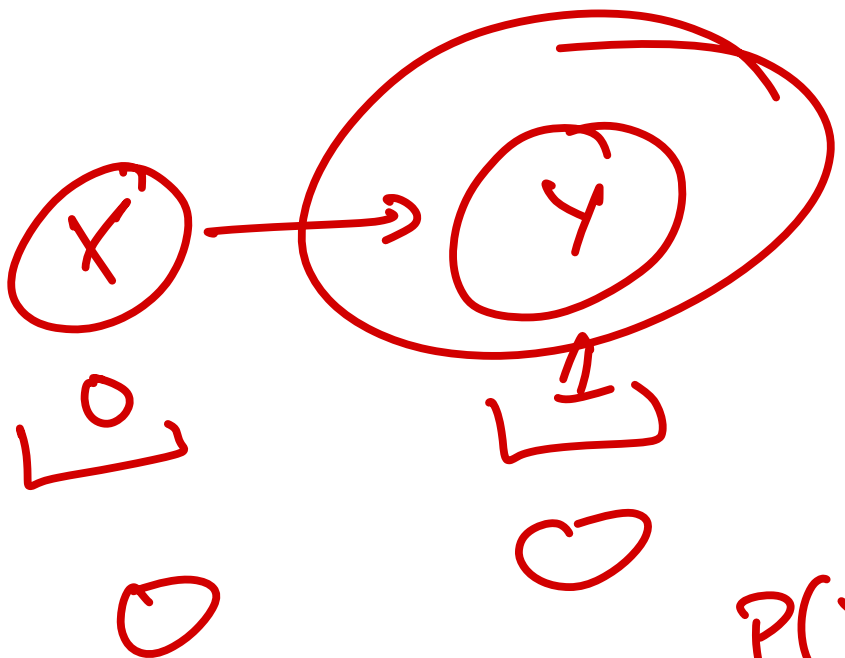
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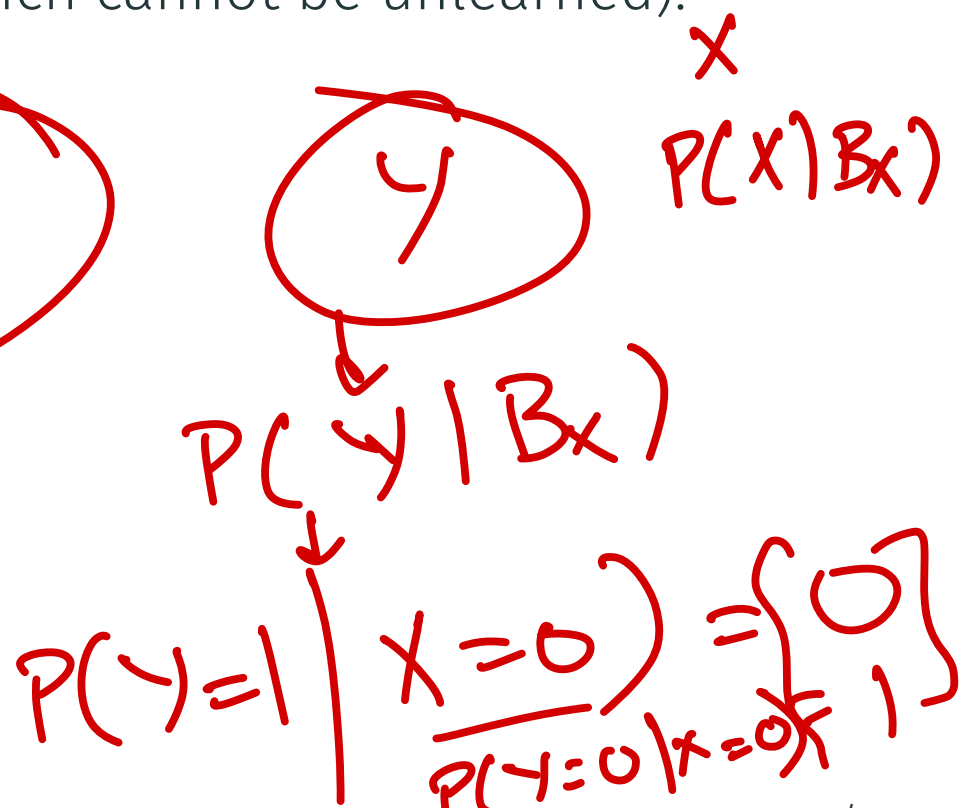
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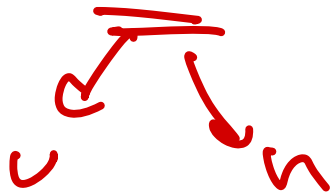
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The # of computations grows linearly in the size of the BN, and also in the amount of data (as expected).

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Note that these are updates ($\longleftarrow$), not equalities ($=$).
The right hand sides depend on the current CPTs.

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*Formulas are great, but what about intuition?*

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EM update

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ML for complete data

$$P(X_i = x) \quad \leftarrow \quad \frac{1}{T}\sum_t P(X_i = x | V_t = v_t)$$

EM update

- At nodes with parents

# Intuition for EM updates — by analogy

- At root nodes

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ML for complete data

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EM update

- Special case

# Intuition for EM updates — by analogy

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- At nodes with parents

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- Special case

  *Consider a CPT whose nodes are fully observed.*

# Intuition for EM updates — by analogy

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EM update

- Special case

*Consider a CPT whose nodes are fully observed.*
*EM updates in this case reduce to ML estimates for complete data.*

# EM updates

# EM updates

$$P(X_i = x) \longleftarrow \frac{1}{T} \sum_t P(X_i = x | V_t = v_t) \quad \text{root nodes}$$

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Intuitively:

# EM updates

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## Intuitively:

When the data is complete, we estimate the CPTs from observed counts.

# EM updates

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## Intuitively:

When the data is complete, we estimate the CPTs from observed counts.

When the data is incomplete, we re-estimate the CPTs from expected counts.

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## Intuitively:

When the data is complete, we estimate the CPTs from observed counts.

When the data is incomplete, we re-estimate the CPTs from expected counts.

These expected counts are computed from the posterior distributions $P(h|v_t)$.

# Key properties of EM

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  The updates do not require the tuning of a learning rate ($\eta > 0$), as in most gradient-based methods.

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- **No learning rate**

  The updates do not require the tuning of a learning rate ($\eta > 0$), as in most gradient-based methods.

- **Monotonic convergence**

  The updated CPTs from EM always increase the incomplete-data log-likelihood $\mathcal{L} = \sum_t \log P(V_t = v_t)$.

# Check In

Q. How much of EM did you understand?

A. (Nearly) All of it

B. Some of it, but I have some doubts

C. Maybe a little, but I'm pretty confused

D. Almost none of it; I'm totally lost



# Log-likelihood

# Log-likelihood

- Incomplete data set

# Log-likelihood

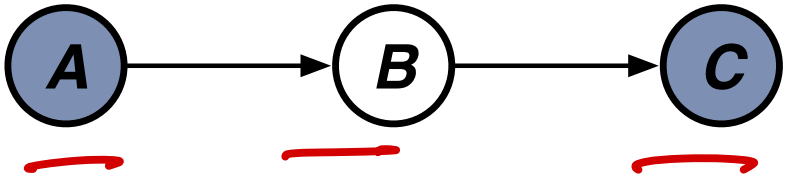
- Incomplete data set

| $t$ | $A$ | $B$ | $C$ |
|---|---|---|---|
| 1 | $a_1$ | ? | $c_1$ |
| 2 | $a_2$ | ? | $c_2$ |
| ⋮ | ⋮ | ⋮ | ⋮ |
| T | $a_T$ | ? | $c_T$ |

# Log-likelihood

- Incomplete data set

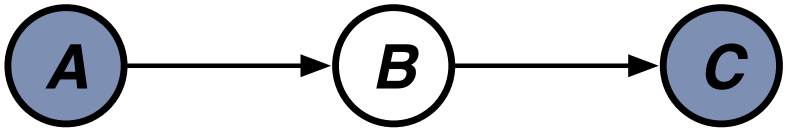
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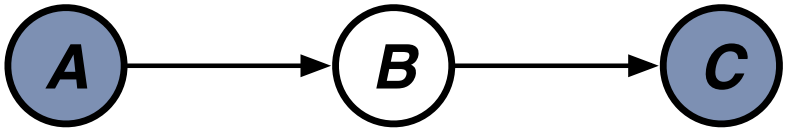

How to choose the CPTs
to maximize the log-likelihood
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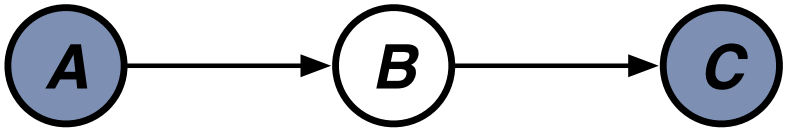



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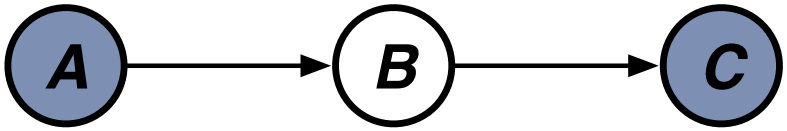
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$$\mathcal{L} = \sum_t \log P(a_t, c_t)$$

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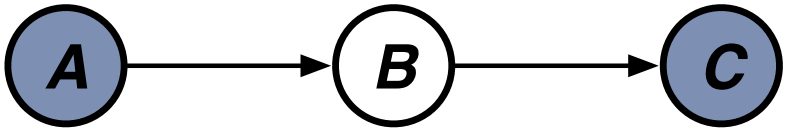
$$= \sum_t \log \sum_b P(a_t, b, c_t)$$

marginalization

# Log-likelihood

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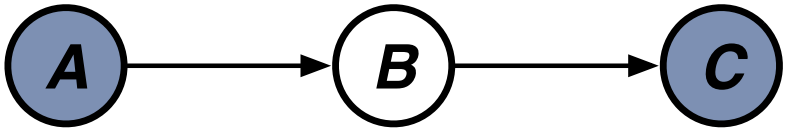
$$= \sum_t \log \sum_b P(a_t, b, c_t) \qquad \text{marginalization}$$

$$= \sum_t \log \sum_b P(a_t)\, P(b|a_t)\, P(c_t|a_t, b) \qquad \text{product rule}$$

# Log-likelihood

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- Log-likelihood

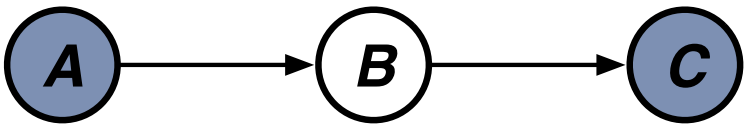
$$\mathcal{L} = \sum_t \log P(a_t, c_t)$$

$$= \sum_t \log \sum_b P(a_t, b, c_t) \qquad \boxed{\text{marginalization}}$$

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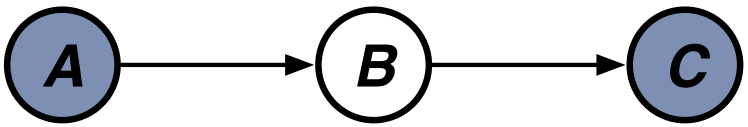
$$= \sum_t \log \sum_b P(a_t)\, P(b|a_t)\, P(c_t|b) \qquad \boxed{\text{conditional independence}}$$

# Example




Suppose that $A$ and $C$ are observed and $B$ is hidden.

# Example




Suppose that $A$ and $C$ are observed and $B$ is hidden.

Q. Which parameters of this network can you estimate directly from the data (in one step—no iteration required)?

A. $P(A)$

B. $P(B|A)$

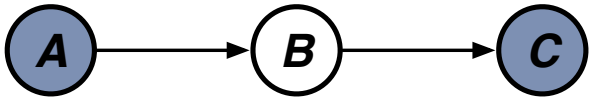
C. $P(C|B)$

D. Both A. and C.

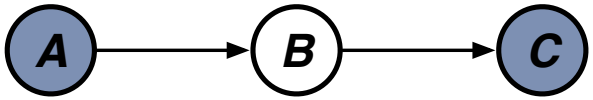
E. None of them

# EM update for $P(A)$

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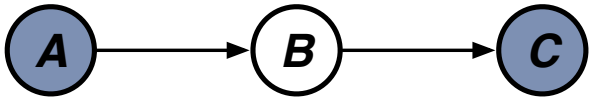


# EM update for $P(A)$




- General form

# EM update for $P(A)$

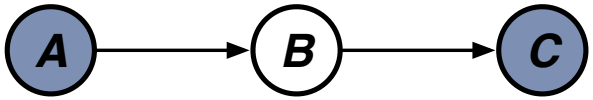



- General form

$$P(X_i = x) \longleftarrow \frac{1}{T} \sum_t P(X_i = x | V_t = v_t)$$

root node

# EM update for $P(A)$



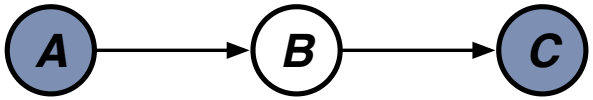

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root node

- Update for this CPT

# EM update for $P(A)$




- General form

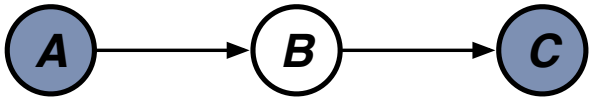
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root node

- Update for this CPT

$$P(A = a)$$

# EM update for $P(A)$




- General form

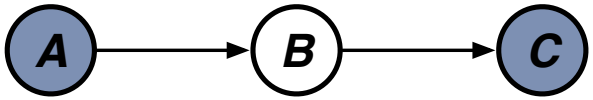
$$P(X_i = x) \longleftarrow \frac{1}{T} \sum_t P(X_i = x | V_t = v_t)$$

root node

- Update for this CPT

$$P(A = a) \longleftarrow \frac{1}{T} \sum_t P(A = a | A = a_t, C = c_t)$$

# EM update for $P(A)$




- General form

$$P(X_i = x) \longleftarrow \frac{1}{T} \sum_t P(X_i = x | V_t = v_t)$$

root node

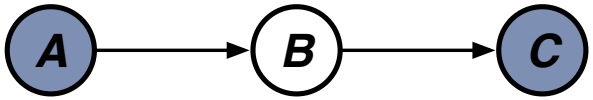
- Update for this CPT

$$P(A = a) \longleftarrow \frac{1}{T} \sum_t P(A = a | A = a_t, C = c_t)$$

Simplify:

$$P(A = a) \longleftarrow$$

# EM update for $P(A)$




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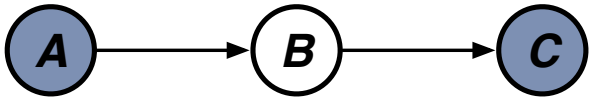
- Update for this CPT

$$P(A = a) \longleftarrow \frac{1}{T} \sum_t P(A = a | A = a_t, C = c_t)$$

Simplify:

$$P(A = a) \longleftarrow \frac{1}{T} \sum_t I(a, a_t)$$

# EM update for $P(A)$




- General form

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root node

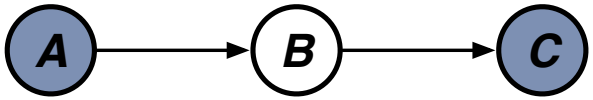
- Update for this CPT

$$P(A = a) \longleftarrow \frac{1}{T} \sum_t P(A = a | A = a_t, C = c_t)$$

Simplify:

$$P(A = a) \longleftarrow \frac{1}{T} \sum_t I(a, a_t) = \frac{1}{T} \text{count}(A = a)$$

# EM update for $P(A)$




- General form

$$P(X_i = x) \leftarrow \frac{1}{T} \sum_t P(X_i = x | V_t = v_t)$$ root node

- Update for this CPT

$$P(A = a) \leftarrow \frac{1}{T} \sum_t P(A = a | A = a_t, C = c_t)$$

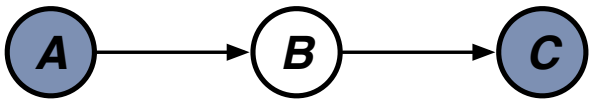
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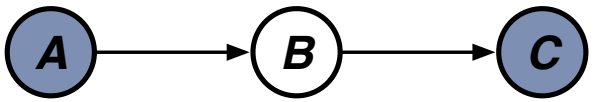
The update reduces to the ML estimate for complete data—as it must, because $A$ is observed and has no unobserved parents.

# EM update for $P(B|A)$

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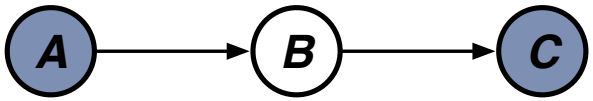


# EM update for $P(B|A)$




- General form

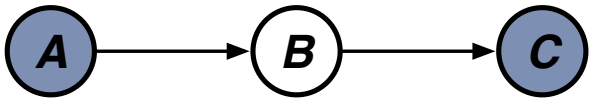
# EM update for $P(B|A)$




- General form

$$P(X_i = x | \mathrm{pa}_i = \pi) \leftarrow \frac{\sum_t P(X_i = x, \mathrm{pa}_i = \pi | V_t = v_t)}{\sum_t P(\mathrm{pa}_i = \pi | V_t = v_t)}$$

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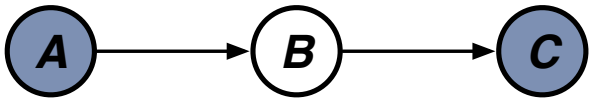



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- Update for this CPT

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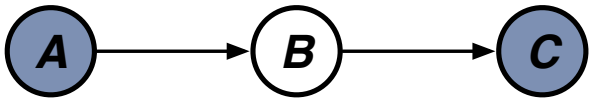

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- Update for this CPT

$$P(B = b | A = a)$$

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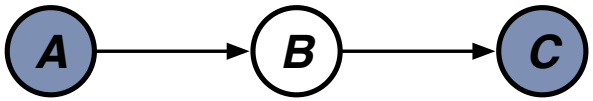

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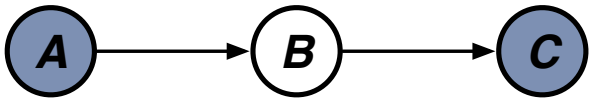

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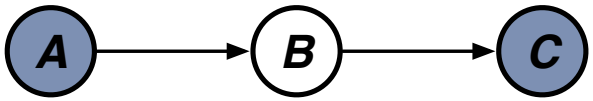
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Simplify:

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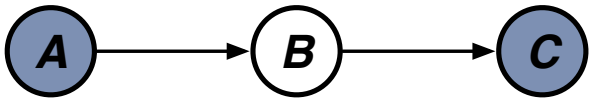
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$$P(B = b|A = a) \longleftarrow \sum_t I(a, a_t) P(B = b|A = a_t, C = c_t)$$

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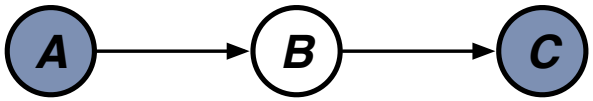
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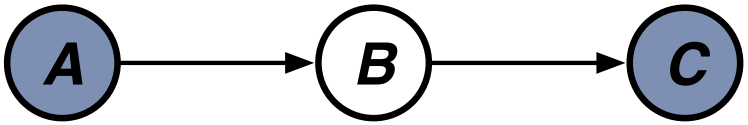
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Simplify:

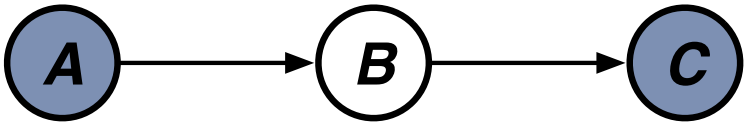
$$P(B = b | A = a) \longleftarrow \frac{\sum_t I(a, a_t) P(B = b | A = a_t, C = c_t)}{\sum_t I(a, a_t)}$$

computed from Bayes rule

# Example



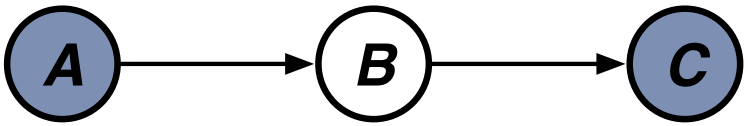
# Example




Suppose that $A$ and $C$ are observed and $B$ is hidden.

- Inference

# Example

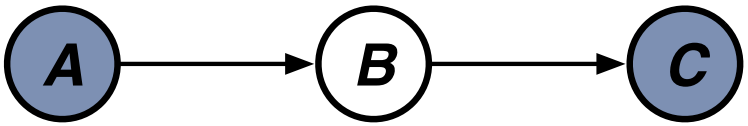



Suppose that $A$ and $C$ are observed and $B$ is hidden.

- Inference

$P(B = b | A = a, C = c)$

# Example

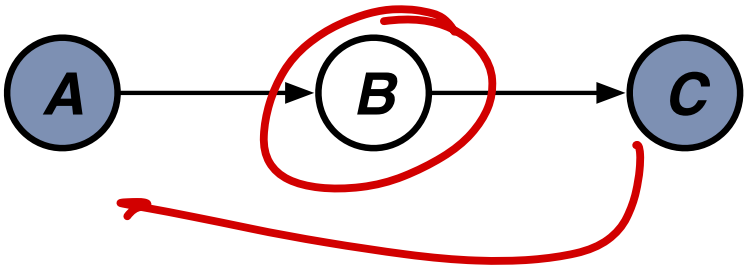



Suppose that $A$ and $C$ are observed and $B$ is hidden.

- Inference

$$P(B=b|A=a,C=c) = \frac{P(C=c|B=b,A=a)\,P(B=b|A=a)}{P(C=c|A=a)} \quad \boxed{\text{BR}}$$

# Example



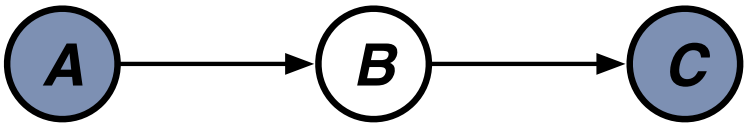

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# Example




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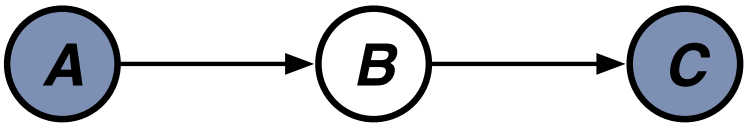
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$$= \frac{P(C=c|B=b)\,P(B=b|A=a)}{P(C=c|A=a)} \quad \boxed{\text{CI}}$$

$$= \frac{P(C=c|B=b)\,P(B=b|A=a)}{\sum_{b'} P(C=c|B=b')\,P(B=b'|A=a)} \quad \boxed{\text{normalized}}$$

# Example




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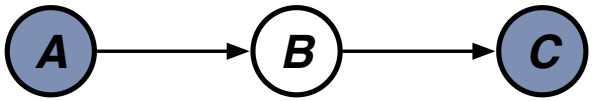
CPT → ↓ CPT

$$= \frac{P(C=c|B=b)\,P(B=b|A=a)}{\sum_{b'} P(C=c|B=b')\,P(B=b'|A=a)} \quad \text{normalized}$$

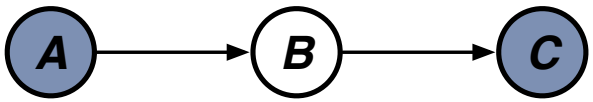
This is the only non-trivial posterior probability that we'll need for the EM updates in this example.

# EM update for $P(C|B)$

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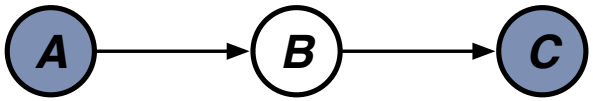


# EM update for $P(C|B)$




- General form

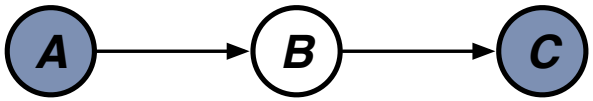
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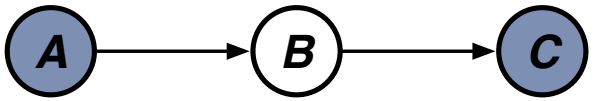



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- Update for this CPT

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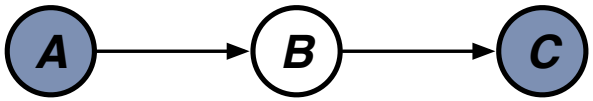

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$$P(C = c | B = b)$$

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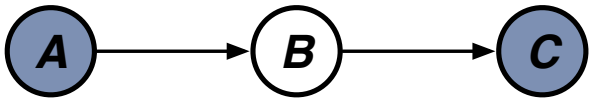

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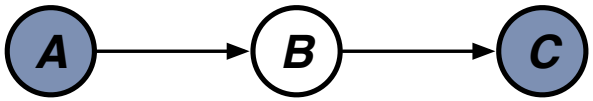

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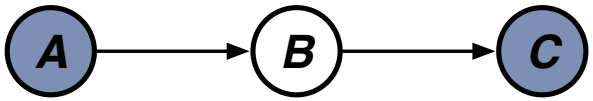
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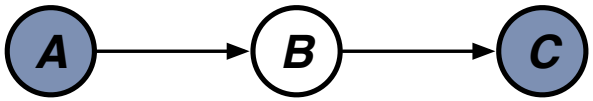
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# Summary of EM algorithm

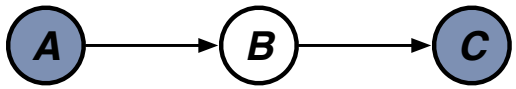
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- **E-step** (Inference)

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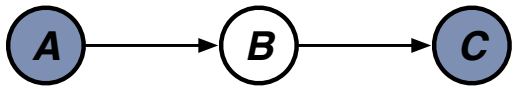
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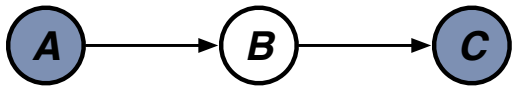



- **M-step** (Learning)

# Summary of EM algorithm

- **E-step** (Inference)

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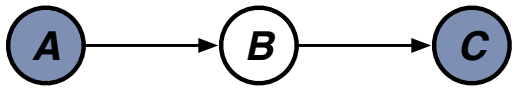

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$$P(a) = \frac{1}{T}\,\mathrm{count}(A=a)$$

# Summary of EM algorithm

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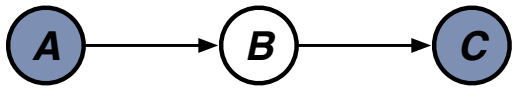
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# Summary of EM algorithm

- **E-step** (Inference)

$$P(b|a_t, c_t) = \frac{P(c_t|b)\,P(b|a_t)}{\sum_{b'} P(c_t|b')\,P(b'|a_t)}$$




- **M-step** (Learning)

$$P(a) = \frac{1}{T}\,\mathrm{count}(A=a)$$

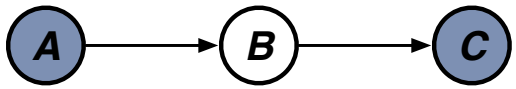
$$P(b|a) \longleftarrow \frac{\sum_t I(a, a_t)\,P(b|a_t, c_t)}{\sum_t I(a, a_t)}$$

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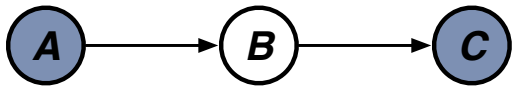
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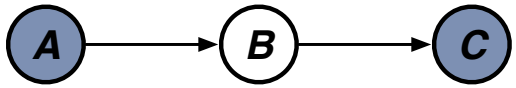
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- **Convergence**

  There are no learning rates to tune.
  Each update increases the incomplete data log-likelihood:

$$\mathcal{L} = \sum_t \log \sum_b P(a_t)\,P(b|a_t)\,P(c_t|b)$$

# Check In

Q. How much of EM did you understand?

A. (Nearly) All of it

B. Some of it, but I have some doubts

C. Maybe a little, but I'm pretty confused

D. Almost none of it; I'm totally lost

# That’s all folks!